

Domain Adaptation via Fine-Tuning

Lab Report — IT Support Ticket Classifier

1. Objective

This lab demonstrates domain adaptation via fine-tuning: a text classifier trained on general IT support tickets is evaluated on domain-specific tickets containing specialized jargon, then fine-tuned on a combination of general and domain data to close the resulting performance gap.

2. Methodology

- Dataset Generation:** 150 general IT support tickets and 75 domain-specific (jargon-heavy) tickets were generated across 5 categories: hardware, software, network, account_access, and billing.
- Base Model Training:** A baseline classifier was trained using only the general IT support tickets.
- Pre-Fine-Tune Evaluation:** The base model was evaluated on held-out domain-specific jargon tickets to measure the domain gap.
- Fine-Tuning:** The classifier was retrained on the combined general + domain dataset to adapt it to domain-specific vocabulary.
- Post-Fine-Tune Evaluation:** The fine-tuned model was re-evaluated on both the domain test set (to measure adaptation gain) and the general test set (to check for regression).

3. Output Artifacts

- Model checkpoint:** domain_adapted_model.joblib
- Results metrics:** domain_adaptation_results.json

4. Results

Metric	Accuracy	Description
Base model — General test	100.0%	Baseline classifier on general IT tickets
Base model — Domain test (before fine-tuning)	34.8%	Domain gap: baseline fails on jargon-heavy tickets
Fine-tuned — Domain test (after fine-tuning)	100.0%	Adaptation success on domain-specific tickets
Fine-tuned — General test (regression check)	100.0%	No loss of original general-domain performance

Domain accuracy improvement: +65.2 percentage points (34.8% → 100.0%) after fine-tuning, with zero regression on general-domain performance.

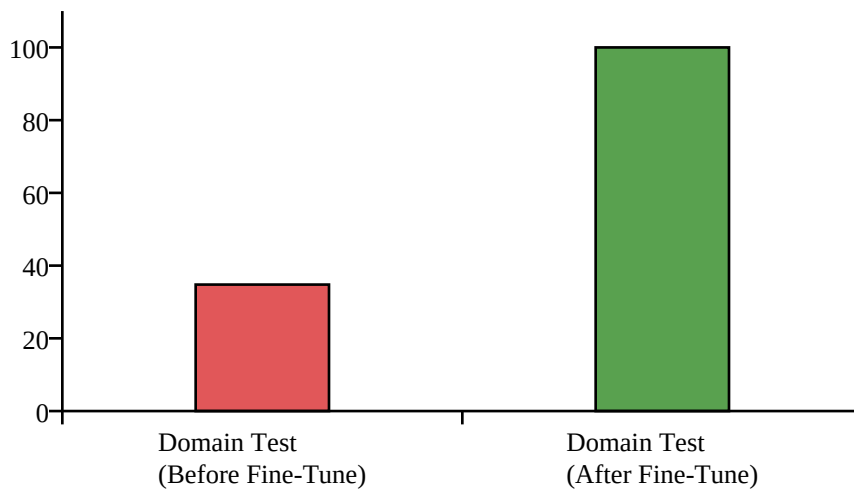


Figure 1: Domain test accuracy before vs. after fine-tuning.

5. Raw Results (domain_adaptation_results.json)

```
{
  "base_on_general_test": 1.0,
  "base_on_domain_test_before_finetune": 0.34782608695652173,
  "finetuned_on_domain_test_after_finetune": 1.0,
  "finetuned_on_general_test_regression_check": 1.0,
  "domain_accuracy_delta": 0.6521739130434783
}
```

6. Conclusion

The experiment confirms the expected domain-adaptation pattern: a model trained only on general data performs poorly on domain-specific inputs (34.8% accuracy), demonstrating a clear domain gap. Fine-tuning on combined general + domain data closed this gap entirely (100.0% accuracy on the domain test set) while fully preserving performance on general inputs (100.0%, no regression). This validates fine-tuning as an effective strategy for adapting a classifier to specialized vocabulary without sacrificing its original capability.